

Quantum Fisher Information for Lorentzian Light-Ray Tomography

Gauge Kernels, Causal Microlocal Visibility, and Finite Null-Channel Design

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ABSTRACT

Linearized propagation through a weak metric perturbation maps a symmetric two-tensor h to the null-ray functional $A_\gamma h = \frac{1}{2} \int_\gamma h_{\mu\nu} k^\mu k^\nu d\lambda$, the symmetric-tensor Lorentzian light-ray transform evaluated on a null geodesic. We place a quantum-statistical layer directly on this transform. A one-photon temporal packet is, at first geometric-optics order, a rigid delay channel with parameter $A_\gamma h$, and a finite network has metric quantum Fisher information (QFI) $J^\top W_\rho J$, where J is the light-ray sensitivity matrix and W_ρ is the QFI of the quantum probe with respect to link delays. Hence $\ker J \subseteq \ker(J^\top W_\rho J)$: quantum resources can redistribute information within a physical forward map but cannot lift a geometric kernel. For independent transform-limited Gaussian photons, W_ρ is diagonal with entries s_a^{-2} and the QFI is the weighted light-ray normal operator.

The universal delay-channel kernel contains linearized diffeomorphisms and conformal perturbations. We introduce a restricted static clock-redshift observable that detects endpoint conformal differences and prove the block-rank law $\text{rank} \begin{pmatrix} D_p & 0 \\ R_p & R_c \end{pmatrix} = d_p + \text{rank } R_c$ whenever the physical delay block D_p has full column rank. In the continuum limit, the Fisher operator is the weighted Lorentzian light-ray normal operator: timelike covectors are microlocally invisible to the normal operator, spacelike covectors are microlocally visible modulo gauge under adequate angular coverage, and lightlike covectors lie on the paired-Lagrangian transition.

We verify the implementation with seven numerical checks and one stop-on-failure process control. A joint delay-redshift experiment across $d_p \in \{6, 8, 12, 16\}$ retains a four-dimensional conformal kernel for the complete 144-ray delay bank, raises rank by three with three clock links,

and removes the modeled conformal nullspace with four. A repository-registered order-1024 campaign covering 64 design cells and five packet-width seeds produces 320 full-rank records, median blocked-sector retention 0.989 with a worst registered cell of 0.188 under one pinned 20% angular exclusion, and millisecond selection against second-scale Jacobian assembly. A post-registration reduction of those same records shows that enlarging the candidate pool from 256 to 4096 rays increases the pool-size-adjusted worst-direction statistic by factors 1.98–4.03, while doubling K/d from 1 to 2 improves the fixed-pool R -normalized minimum eigenvalue by factors 2.39–3.19 without a rank failure. Submission controls show that QFI-aware ray selection improves the true weighted minimum eigenvalue by median factors 2.32–7.18 over weight-blind selection on three representative cells, while the unquotiented Fisher matrix retains nullity four at every tested ray budget. A one-shot DeepSets selector beats random and angular baselines, but its median operating point is Pareto dominated by greedy Fisher-aware design in both objective value and latency, providing an explicit negative result for one-shot, selection-history-independent top- K scoring. The executed finite-design experiments use independent probes; correlated-probe advantage is deferred to the companion study. The framework links Lorentzian tensor tomography, distributed quantum sensing, statistical inverse problems, and finite experimental design while separating exact identifiability from finite-noise precision.

Keywords: Lorentzian light-ray transform; quantum Fisher information; tensor tomography; gauge kernel; conformal invariance; microlocal analysis; optimal experimental design; quantum metrology; clock redshift; distributed quantum sensing.

1 Introduction

Light propagation is simultaneously a geometric probe and a quantum communication channel. In weak gravity, the leading perturbations of optical phase, travel time, redshift, and deflection are linear functionals of the metric perturbation. These functionals are not arbitrary: they are light-ray transforms of a symmetric two-tensor, together with endpoint and derivative terms.

Separately, quantum metrology equips a parameterized family of output states with a Fisher metric. The central observation of this paper is that these structures compose exactly.

If a network of photon or clock channels depends on a finite metric parameter vector θ through link parameters $\mathbf{x} = \mathbf{J}\theta$, and the quantum probe has link-parameter QFI $\mathbf{W}_\rho \succeq \mathbf{0}$, then the metric QFI is

$$\mathbf{F}_\theta = \mathbf{J}^\top \mathbf{W}_\rho \mathbf{J}. \quad (1)$$

This elementary pullback has geometric consequences. The forward-map kernel is an absolute statistical nullspace; a positive quantum weight can improve precision only on the image of $\mathbf{J}$. In the continuum, the same construction becomes the weighted normal operator of Lorentzian integral geometry. Packet bandwidth, photon allocation, entanglement covariance, angular coverage, gauge, and microlocal ill-posedness therefore enter a single operator framework.

A second distinction is equally important. Adding more measurements through a channel cannot reveal a direction annihilated by that channel. Adding a different physical observable can. Delay rays satisfy $\mathbf{A}_\gamma(\phi\eta) = \mathbf{0}$ pointwise for every conformal perturbation. Static endpoint-frequency comparisons instead measure differences of $\mathbf{h}_{00}$ and can restore selected conformal contrasts. This distinction is both a theorem and the central computational result of the paper.

1.1 Relation to prior work

The Lorentzian light-ray transform has a mature inverse-problem theory. Lassas, Oksanen, Stefanov, and Uhlmann analyzed the scalar transform as a Fourier integral operator and showed recovery of spacelike singularities under suitable hypotheses [1]. Wang constructed parametrices in Minkowski spacetime and described the global normal operator by paired-Lagrangian rather than globally elliptic pseudodifferential structure [2, 3]. Tensor transforms possess the natural potential and conformal gauge; injectivity modulo that gauge is known in several Minkowski, static, and stationary settings [4–7]. These results identify what ideal geometric data can recover.

A statistical layer already exists for Riemannian X-ray tomography. Monard, Nickl, and Paternain identify I^*I as the Fisher-information operator governing efficiency and Bernstein–von Mises limits [8], and adaptive Bayesian designs have been studied for conventional X-ray imaging [9]. These works are classical, Riemannian, and primarily scalar.

Quantum-metrology studies have independently derived quantum Cramér–Rao bounds for parameters of classical spacetimes [10]; photon packets have been used to estimate Schwarzschild and terrestrial parameters through gravitational redshift and mode deformation [11, 12]; and detector-based programs recover spacetime intervals and metric information from local quantum measurements [13]. Those works typically begin with a specified low-dimensional metric family. They do not characterize the full potential–conformal delay-channel kernel, the Lorentzian tensor normal operator, or finite null-channel design on the gauge quotient.

Distributed quantum sensing provides the appropriate network-metrology canon. General results characterize when entanglement and collective measurements improve estimation of global or local functions of sensor parameters [14, 15]; continuous-variable multipartite protocols realize analogous network advantages [16]; entangled optical-clock networks were proposed by Kómár and collaborators [17]; and the standard-versus-Heisenberg scaling framework is reviewed by Giovannetti, Lloyd, and Maccone [18]. The product–GHZ tradeoff used below is therefore established distributed-sensing theory. The contribution here is its composition with a Lorentzian tensor sensitivity map J , whose geometric kernel and microlocal image are derived rather than assumed.

The targeted contribution is therefore not that Fisher information has never appeared beside a ray transform, nor that quantum probes of gravity are new. It is the combined framework

Lorentzian symmetric-tensor light-ray tomography on the potential–conformal quotient, with photon-channel QFI, causal microlocal visibility, clock-assisted kernel lifting, and finite-network design.

1.2 Contributions

The paper makes eight contributions.

1. We derive the weak-field null-delay functional from the eikonal equation and show that its first geometric-optics correction is achromatic.
2. We prove the finite-channel QFI factorization and its arbitrary-probe extension. The resulting quantum non-lifting principle states $\ker J \subseteq \ker(J^\top W_\rho J)$.
3. We separate the universal potential-conformal delay kernel from the larger nullspace created by finite and incomplete ray sampling.
4. We introduce a restricted static endpoint-redshift channel and prove clock-assisted conformal rank restoration, including the block-rank formula with a nonzero physical redshift block R_p .
5. We identify the continuum QFI with the weighted Lorentzian light-ray normal operator and interpret its spacelike, timelike, and lightlike microlocal regimes statistically.
6. We derive the exact one-shot Helstrom error for distinguishing two weak geometries with one Gaussian photon.
7. We release numerical checks for the analytic identities and a registered deterministic campaign covering 64 design cells, five packet-width seeds, heterogeneous photon bandwidths, angular exclusions, and both raw and R -normalized posterior conventions; a post-registration reduction quantifies candidate-density and ray-budget returns without rerunning the forward model.
8. We report an honest negative ML result: a repaired one-shot DeepSets selector beats weak baselines, but its median operating point is dominated in both quality and latency by an optimized deterministic greedy method.

1.3 Scope and evidentiary status

The primary model is linearized, lossless, and geometric-optical. Metric perturbations are smooth and compactly supported in an interior world tube unless endpoint observables are introduced explicitly. The finite-ray adjoint is distributional; the continuum ray measure is the setting in which an ordinary inverse-problem normal operator emerges. The static redshift section fixes static emitter and receiver worldlines and does not claim a complete moving-detector gauge treatment. Numerical checks validate implementation paths but do not replace proof. Pulsar timing and interferometric applications are discussed only as a separate lightlike, on-shell, endpoint-sensitive program. The metric

families used in the finite experiments are kinematic test families and are not constrained to satisfy the linearized Einstein equations or a specified source model. Imposing on-shell field equations may remove, couple, or reparameterize some of these directions, and that analysis is deferred to detector-facing studies.

Repository registration and execution chronology. The registration artifact is `configs/experiment.yaml`. Source snapshot `ee28bd9`, timestamped 18 August 2026 at 21:32:41 UTC, recorded the design surface, seed set, metrics, acceptance gates, and planned ablations. It predates the first registered-seed scaling execution, snapshot `5094132` at 22:25:35 UTC, and the full-metric execution, snapshot `bf4002e` on 19 August 2026 at 00:30:12 UTC. Corrective and supplementary controls are identified by later snapshots, including `a384312` at 01:39:06 UTC. The density–budget tables are an analysis-only reduction of the already executed campaign, committed as `fdd6cfa` at 02:27:45 UTC; they introduce no new forward numerics. The accompanying `REGISTRATION.md` records the full hashes, UTC timestamps, artifact hashes, and which decisions were fixed before each execution. Throughout, “registered” means this commit-pinned repository registration; it is not a claim of third-party preregistration. No archival DOI is claimed in this working-paper version; a DOI-minted repository deposit should replace the development-repository link before journal submission.

2 Weak-field null channels

2.1 Background and ray parameterization

Let $M = \mathbb{R}^{1+n}$, $n \geq 3$, with coordinates $x = (t, \mathbf{x})$ and Minkowski metric $\eta = \text{diag}(-1, 1, \dots, 1)$. Let $\Omega \Subset M$ be a compact interior world tube and $\mathcal{E} = C_c^\infty(\Omega; S^2 T^* M)$. The restriction $n \geq 3$ matches the symmetric-two-tensor kernel and microlocal results invoked below, including the physical $1 + 3$ dimensional case. The elementary eikonal and QFI identities themselves do not require this restriction. Consider a weak metric family

$$g_\varepsilon = \eta + \varepsilon h + O(\varepsilon^2), \quad h \in \mathcal{E}. \quad (2)$$

For $\vartheta \in S^{n-1}$ define the null vector $k_\vartheta = (1, \vartheta)$, $\eta(k_\vartheta, k_\vartheta) = 0$, and the affine background ray

$$\gamma_{y,\vartheta}(\lambda) = (\lambda, y + \lambda\vartheta), \quad y \in \mathbb{R}^n. \quad (3)$$

Every complete future-directed null line with tangent proportional to $(1, \vartheta)$ intersects the hypersurface $t = 0$ once, so y is its unrestricted spatial intercept. The finite numerical candidate banks later use a transverse-offset subfamily with $y \cdot \vartheta = 0$; that is an experimental-design restriction, not the parameterization of the complete continuum ray manifold. The symmetric two-tensor light-ray transform is

$$(L_2 h)(y, \vartheta) = \int_{\mathbb{R}} h_{\mu\nu}(\gamma_{y,\vartheta}(\lambda)) k_{\vartheta}^{\mu} k_{\vartheta}^{\nu} d\lambda. \quad (4)$$

Compact support makes the integral finite.

2.2 Eikonal linearization

Lemma 2.1 (*Linearized eikonal delay*). Let $S_{\varepsilon} = S_0 + \varepsilon S_1 + O(\varepsilon^2)$ solve the eikonal equation $g_{\varepsilon}^{\mu\nu} \partial_{\mu} S_{\varepsilon} \partial_{\nu} S_{\varepsilon} = 0$, and let $k^{\mu} = \eta^{\mu\nu} \partial_{\nu} S_0$ generate an affine background null ray γ . Then

$$\frac{d}{d\lambda} S_1(\gamma(\lambda)) = \frac{1}{2} h_{\mu\nu}(\gamma(\lambda)) k^{\mu} k^{\nu}. \quad (5)$$

With the phase convention used here, the first-order delay parameter is

$$A_{\gamma} h := \frac{1}{2} \int_{\gamma} h_{\mu\nu} k^{\mu} k^{\nu} d\lambda = \frac{1}{2} (L_2 h)(\gamma). \quad (6)$$

The opposite global sign convention yields the same Fisher information.

Proof. The inverse metric satisfies $g_{\varepsilon}^{\mu\nu} = \eta^{\mu\nu} - \varepsilon h^{\mu\nu} + O(\varepsilon^2)$. At order ε the eikonal equation becomes

$$2\eta^{\mu\nu} (\partial_{\mu} S_0) (\partial_{\nu} S_1) - h^{\mu\nu} (\partial_{\mu} S_0) (\partial_{\nu} S_0) = 0.$$

Since $k^{\mu} = \eta^{\mu\nu} \partial_{\nu} S_0$ and $h^{\mu\nu} \partial_{\mu} S_0 \partial_{\nu} S_0 = h_{\alpha\beta} k^{\alpha} k^{\beta}$, we obtain $2k^{\mu} \partial_{\mu} S_1 = h_{\mu\nu} k^{\mu} k^{\nu}$. Along an affine integral curve of k , $k^{\mu} \partial_{\mu} = d/d\lambda$, giving (5) and (6). $\square$

Remark 2.2 (*Achromatic leading order*). The functional $A_{\gamma} h$ contains no probe frequency. To first geometric-optics order every spectral component receives the same coordinate delay, so the wave packet undergoes a rigid translation. Dispersion, frequency-dependent matter response, diffraction, path-dependent loss, and beyond-eikonal corrections mark the boundary of the model.

The same light-ray object also organizes neighboring observables. Deflection is obtained from transverse derivatives of the integrated phase, while frequency shifts introduce endpoint and derivative terms. The present paper isolates delay first and introduces a controlled static redshift channel in Section 5.

3 Quantum Fisher information of photon delay channels

3.1 One-photon temporal modes

Let $\psi \in L^2(\mathbb{R}, dt)$ be a normalized temporal mode with finite frequency variance. The corresponding one-photon state is $|\psi\rangle = a^\dagger(\psi)|0\rangle$. Let $\hat{\nu} = -i\partial_t$ denote the baseband frequency generator. A delay τ acts through $U_\tau = e^{-i\tau\hat{\nu}}$, $|\psi_\tau\rangle = U_\tau|\psi\rangle$. A source clock, local oscillator, interferometric reference arm, or equivalent timing reference is assumed. Without a relational reference, a common translation of every relevant degree of freedom is not observable.

For a metric perturbation h , define

$$|\psi_h\rangle = e^{-i(A_\gamma h)\hat{\nu}}|\psi\rangle. \quad (7)$$

The QFI is naturally a bilinear form on perturbation directions $u, v \in \mathcal{E}$ [19, 20].

Proposition 3.1 (*Single-channel QFI*). *For the pure-state family (7),*

$$F_\gamma(u, v) = 4 \text{Var}_\psi(\hat{\nu}) (A_\gamma u)(A_\gamma v). \quad (8)$$

Consequently, a single scalar delay channel has rank one on every finite-dimensional metric family on which A_γ is nonzero.

Proof. The directional derivative is $D_u|\psi_h\rangle = -i(A_\gamma u)\hat{\nu}|\psi_h\rangle$. For a pure state, the QFI bilinear form is $4 \text{Re}(\langle D_u\psi_h | D_v\psi_h \rangle - \langle D_u\psi_h | \psi_h \rangle \langle \psi_h | D_v\psi_h \rangle)$. Substitution gives (8). $\square$

Corollary 3.2 (*Transform-limited Gaussian*). *Let*

$$\psi_{\tau,s}(t) = (2\pi s^2)^{-1/4} \exp \left[-\frac{(t - \tau)^2}{4s^2} \right]. \quad (9)$$

Then $\text{Var}(\hat{\nu}) = 1/(4s^2)$ and

$$F_\gamma(u, v) = \frac{1}{s^2} (A_\gamma u)(A_\gamma v). \quad (10)$$

The ideal time-of-arrival density is $\mathcal{N}(A_\gamma h, s^2)$, whose classical Fisher information equals (10); time-of-arrival detection therefore attains the QFI in this ideal shift family.

The transform-limited qualifier is essential. A chirped packet can have the same arrival-time intensity width but larger spectral variance. Its QFI is then larger than $1/s^2$, while direct time-of-arrival measurement need not attain it.

3.2 Finite networks

Let $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ be a finite set of rays and define

$$A_\Gamma : \mathcal{E} \rightarrow \mathbb{R}^m, \quad (A_\Gamma h)_a = A_{\gamma_a} h. \quad (11)$$

For independent one-photon probes let $W = \text{diag}(w_1, \dots, w_m)$, $w_a = 4 \text{Var}_{\psi_a}(\hat{v}_a) > 0$. For Gaussian packets, $w_a = s_a^{-2}$.

Theorem 3.3 (*Finite-channel quantum Fisher factorization (Theorem A)*). *On the test-tensor space $\mathcal{E}$, the product-state QFI is*

$$F_\Gamma(u, v) = \langle A_\Gamma u, W A_\Gamma v \rangle_{\mathbb{R}^m}. \quad (12)$$

It is positive semidefinite and, because $W \succ 0$,

$$\text{rad } F_\Gamma = \ker A_\Gamma. \quad (13)$$

Its associated finite-rank distributional operator is

$$\mathcal{N}_\Gamma^Q = A_\Gamma^* W A_\Gamma. \quad (14)$$

For every finite-dimensional subspace $V \subset \mathcal{E}$,

$$\text{rank}(F_\Gamma|_V) = \text{rank}(A_\Gamma|_V) \leq m. \quad (15)$$

Proof. QFI adds for independent product probes, and Proposition 3.1 gives (12). Since $F_\Gamma(u, u) = \|W^{1/2} A_\Gamma u\|_2^2$, positivity and (13) follow. The adjoint is taken in the distributional pairing between test tensors and compactly supported tensor distributions. On a finite-dimensional subspace, multiplication by invertible $W^{1/2}$ preserves rank, proving (15). $\square$

For a fixed ray, A_γ is not continuous on an unweighted spacetime L^2 space: it samples a measure-zero submanifold. Explicitly,

$$A_\Gamma^* c = \frac{1}{2} \sum_{a=1}^m c_a k_a^{\otimes 2} \delta_{\gamma_a}, \quad (16)$$

up to the chosen tensor-density convention. Thus (14) is a sum of ray-supported rank-one distributions. Averaging over a continuum of rays produces the conventional normal operator studied in Section 6.

Remark 3.4 (*Correlated pure probes*). For a pure multimode state with commuting delay generators, W generalizes to $W_{ab} = 4 \text{Cov}(\hat{v}_a, \hat{v}_b) \succeq 0$. Then $\text{rad } F_\Gamma = \ker(W^{1/2} A_\Gamma) \supseteq \ker A_\Gamma$. Equality requires positive definiteness on the used generator subspace. The main theorem keeps independent probes so that $W \succ 0$ is explicit.

3.3 Arbitrary quantum probes and the non-lifting principle

Let a finite metric family be written $h_\theta = \sum_{j=1}^d \theta_j e_j$, and let $J_{aj} = A_{\gamma_a} e_j$ be its link-delay Jacobian. Suppose a possibly correlated or entangled probe state ρ produces a QFI matrix $W_\rho \succeq 0$ for the intermediate link parameters $x = J\theta$. For pure probes with commuting generators G_a , $(W_\rho)_{ab} = 4 \text{Cov}_\rho(G_a, G_b)$.

Proposition 3.5 (*Arbitrary-probe pullback and quantum non-lifting*). *The metric-parameter QFI is $F_\theta = J^\top W_\rho J$. Moreover,*

$$\ker J \subseteq \ker F_\theta, \quad \text{rank } F_\theta \leq \min\{\text{rank } J, \text{rank } W_\rho\}. \quad (18)$$

If W_ρ is positive definite on $\text{Im } J$, then $\ker F_\theta = \ker J$.

Proof. For any direction v , $v^\top F_\theta v = (Jv)^\top W_\rho (Jv) = \|W_\rho^{1/2} Jv\|_2^2$. Thus $\ker F_\theta = \ker(W_\rho^{1/2} J)$ and the stated inclusion and rank bound follow. Positive definiteness on $\text{Im } J$ gives equality of kernels. $\square$

The pullback (1) is the pullback of the SLD quantum Fisher matrix. For general correlated multiparameter probes it is an upper-bound information geometry: simultaneous attainability requires a compatibility or Holevo-bound analysis [23]. The numerical campaigns in this paper use independent Gaussian shift channels, for which product time-of-arrival measurements attain the stated QFI.

If the parameter encoding is $U_\theta = \exp[-i \sum_a (J\theta)_a G_a]$ with commuting generators, then $J(\theta_1 - \theta_0) = \mathbf{0}$ implies identical channel unitaries. No ancilla, entangled input, adaptive control, or collective measurement can distinguish such parameter values through that channel architecture. Quantum resources may improve or rebalance visible directions, but they cannot manufacture a missing observable.

Remark 3.6 (*Product versus GHZ probes*). As is well known in distributed quantum sensing [14–18], entanglement advantage can attach to a selected global combination rather than to every local parameter. For m qubit phase sensors with $G_a = Z_a/2$, the product probe $|+\rangle^{\otimes m}$ gives $W = I_m$ and $F = J^\top J$. The global GHZ probe gives $W = \mathbf{1}\mathbf{1}^\top$ and $F = (J^\top \mathbf{1})(J^\top \mathbf{1})^\top$, which has rank at most one. The GHZ state raises the QFI for one common phase from m to m^2 , but can collapse multiparameter identifiability. Entanglement should therefore be matched to the visible singular directions of J , not treated as uniformly beneficial. The observation is not a new distributed-sensing tradeoff; its role here is to show how an established probe-state effect is filtered through the geometry-specific map J .

4 Gauge kernel and finite-sampling nullspace

The light-ray transform of a metric perturbation cannot be injective on raw symmetric tensors. The universal obstruction is the linearization of diffeomorphism and conformal invariance. Let $(d_s V)_{\mu\nu} = \frac{1}{2}(\partial_\mu V_\nu + \partial_\nu V_\mu)$ and define

$$\mathcal{G} = \{2d_s V + \phi\eta : V \in C_c^\infty(\Omega; T^*M), \phi \in C_c^\infty(\Omega)\}. \quad (19)$$

Proposition 4.1 (*Potential and conformal kernel*). *Let $\gamma : [a, b] \rightarrow \mathbb{R}^{1+n}$ be a finite affine Minkowski null segment with tangent k . Then*

$$A_\gamma(\phi\eta) = 0, \quad (20)$$

$$A_\gamma(2d_s V) = V_\mu(\gamma(b))k^\mu - V_\mu(\gamma(a))k^\mu. \quad (21)$$

For a complete affine ray, the same identity follows by taking $a \rightarrow -\infty$ and $b \rightarrow +\infty$ whenever the limits exist. In particular, compact interior support makes both endpoint values zero. Hence

$$\mathcal{G} \subseteq \ker A_\Gamma = \text{rad } F_\Gamma \quad (22)$$

for every finite network of such rays or segments with vanishing endpoint gauge data.

Proof. Because $\mathbf{k}$ is null, $(\phi\eta)_{\mu\nu}k^\mu k^\nu = \phi\eta(\mathbf{k}, \mathbf{k}) = 0$ pointwise. Along an affine ray, $(2d_s V)_{\mu\nu}k^\mu k^\nu = 2k^\mu k^\nu \partial_\mu V_\nu = 2\frac{d}{d\lambda}(V_\nu k^\nu)$. The factor $1/2$ in (6) gives (21). Compact support away from the endpoints gives zero. $\square$

The endpoint formula is not a technical nuisance; it is the exact failure mode of a coordinate-only travel-time observable. If the generator V moves or reparametrizes the emitter or receiver, a bare path integral is not gauge invariant. One must either impose the interior-support condition or include the endpoint and worldline contributions of the complete observable. The computational protocol tests both the cancellation and its controlled failure.

Define the quotient $\mathcal{Q} = \mathcal{E}/\mathcal{G}$. The finite ray map descends to $A_\Gamma : \mathcal{Q} \rightarrow \mathbb{R}^m$. The remaining nullspace $\ker A_\Gamma$ is the sampling nullspace. It includes perturbations supported away from all selected rays, cancellations of $h(\mathbf{k}, \mathbf{k})$ along the sampled paths, and tensor polarizations not separated by the available directions. In complete-data settings, the proved kernel under the relevant hypotheses reduces to potential plus conformal gauge [4–7]. Finite experiments are categorically less informative.

A terminological caution is required before the clock section. Throughout this paper, “conformal gauge” refers to the natural equivalence class of the delay-only tensor light-ray transform. Only the potential component $2d_s V$ is a linearized coordinate gauge of the metric. A conformal perturbation $\phi\eta$ is a physical metric direction that is null for the delay observable, because $\eta(\mathbf{k}, \mathbf{k}) = 0$ on null rays, but it may become identifiable when the observable set is enlarged. Where the distinction matters below, $\mathcal{G}$ is called the potential–conformal transform kernel or the delay-channel ambiguity rather than a gauge redundancy of Einstein gravity.

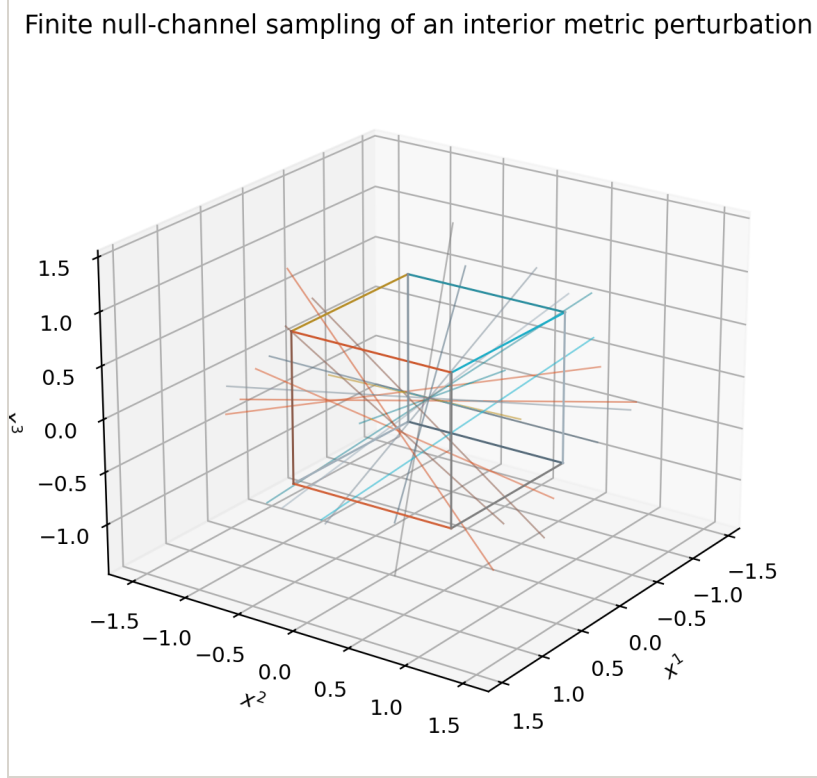
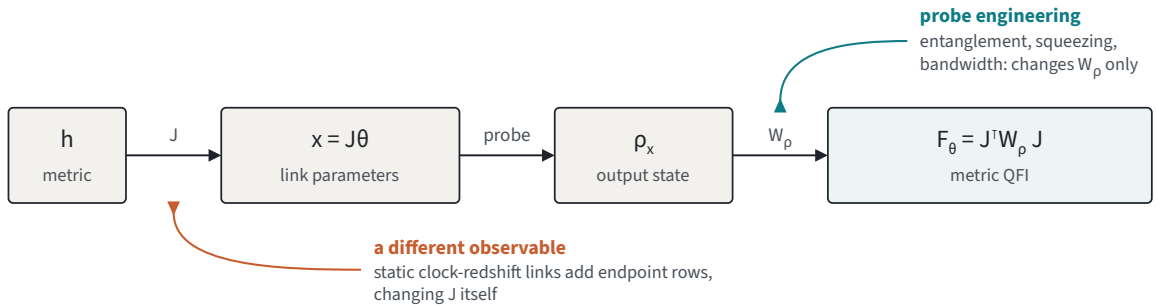


Figure 1: Spatial projection of a finite null-ray network traversing a compact interior world tube. A finite set samples only a small part of the complete light-ray transform and therefore adds a design-dependent nullspace beyond the universal potential and conformal gauge.



$\ker J \subseteq \ker F_\theta$ for every probe state

$\text{rank } F_\theta = d_p + \text{rank } R_c$ once clocks are added

Figure 2: Where quantum resources act, and where they cannot. The metric perturbation reaches the estimator only through the link parameters $x = J\theta$. Probe engineering acts on W_ρ and can redistribute precision across the visible directions, but it leaves $\ker J$ intact (Proposition 3.5). Introducing a different observable changes J itself: static clock-redshift rows add the endpoint block R_c and lift exactly the rank that block carries (Proposition 5.3).

5 Static redshift and conformal rank restoration

The interior-support hypothesis makes the delay theorem exact but also removes endpoint conformal information. This section relaxes that assumption in a restricted static model and introduces operational clocks explicitly.

5.1 Static endpoint functional

Let

$$g_\varepsilon = -N_\varepsilon(\mathbf{x})^2 dt^2 + q_{\varepsilon,ij}(\mathbf{x}) dx^i dx^j, \quad g_\varepsilon = \eta + \varepsilon h + O(\varepsilon^2), \quad (24)$$

with $h_{0i} = 0$ and $\partial_t h = 0$. Static observers have four-velocity $u = N^{-1} \partial_t$. For a null geodesic, $E = -g(k, \partial_t)$ is conserved and the locally measured frequency is $\omega = -g(k, u) = E/N$. For static observers at A and B ,

$$\zeta_{AB}(g) := \log \frac{\omega_B}{\omega_A} = \log N(A) - \log N(B). \quad (25)$$

Since $-g_{00} = 1 - \varepsilon h_{00} + O(\varepsilon^2)$,

$$R_{AB}h := \left. \frac{d}{d\varepsilon} \zeta_{AB}(g_\varepsilon) \right|_0 = \frac{1}{2} (h_{00}(B) - h_{00}(A)). \quad (26)$$

For $h = \phi\eta$,

$$R_{AB}(\phi\eta) = \frac{1}{2} (\phi(A) - \phi(B)). \quad (27)$$

Thus delay rays annihilate every conformal perturbation, while a clock-referenced redshift detects endpoint conformal differences.

5.2 Log-frequency quantum channel

A lossless redshift acts as a dilation on positive frequency. Introduce log-frequency $x = \log(\omega/\omega_*)$ and the unitarily equivalent wavefunction $\chi(x) = \sqrt{\omega} \varphi(\omega)$ in $L^2(\mathbb{R}, dx)$. The redshift ζ becomes the rigid translation $\chi(x) \mapsto \chi(x - \zeta)$, generated by $\hat{p}_x = -i\partial_x$.

Proposition 5.1 (Static redshift QFI). *For a pure log-frequency mode with finite $\text{Var}(\hat{p}_x)$,*

$$F_{AB}^{\text{red}}(u, v) = 4 \text{Var}(\hat{p}_x)(R_{AB}u)(R_{AB}v). \quad (28)$$

For a transform-limited Gaussian in $\mathbf{x}$ with RMS width $\mathbf{s}_x$, the weight is $\mathbf{s}_x^{-2}$.

The proof is identical to Proposition 3.1 after replacing temporal translations by log-frequency translations.

Let $\mathbf{R}$ collect $\mathbf{r}$ static endpoint pairs. Under the independent-block model used in the experiments, let $\mathbf{W}_D \succ \mathbf{0}$ and $\mathbf{W}_R \succ \mathbf{0}$ be the delay and log-frequency QFI matrices. Then

$$\mathcal{N}_{\text{comb}}^Q = \mathbf{A}_\Gamma^* \mathbf{W}_D \mathbf{A}_\Gamma + \mathbf{R}^* \mathbf{W}_R \mathbf{R}. \quad (29)$$

More generally, shared clocks, oscillators, or phase references can correlate the two blocks. Writing $\mathbf{J}_{DR} = (\mathbf{A}_\Gamma, \mathbf{R})^\top$ and allowing an arbitrary joint probe QFI $\mathbf{W}_\rho \succeq \mathbf{0}$ gives $\mathcal{N}_{\text{comb}}^Q = \mathbf{J}_{DR}^* \mathbf{W}_\rho \mathbf{J}_{DR}$, with off-diagonal delay–redshift covariance retained explicitly.

Corollary 5.2 (Clock-assisted conformal rank restoration (Corollary A1)). Let $\mathcal{C} = \text{span}\{\phi_1\eta, \dots, \phi_q\eta\}$ be a finite conformal family and define the endpoint matrix

$$\mathbf{E}_{\ell j} = \frac{1}{2} (\phi_j(\mathbf{A}_\ell) - \phi_j(\mathbf{B}_\ell)). \quad (30)$$

Then delay-only QFI vanishes on $\mathcal{C}$, while the combined QFI restricted to $\mathcal{C}$ is

$$\mathbf{F}_\mathcal{C} = \mathbf{E}^\top \mathbf{W}_R \mathbf{E}, \quad \text{rank } \mathbf{F}_\mathcal{C} = \text{rank } \mathbf{E}. \quad (31)$$

In particular, q independent endpoint comparisons identify the entire family whenever $\mathbf{E}$ has full column rank.

Proposition 5.3 (Block-rank restoration). Let $\mathcal{P}$ be a d_p -dimensional physical family and $\mathcal{C}$ a q -dimensional conformal family. Suppose selected delay and static-redshift observations have the block Jacobian

$$\mathbf{J}_{\text{joint}} = \begin{pmatrix} \mathbf{D}_p & \mathbf{0} \\ \mathbf{R}_p & \mathbf{R}_c \end{pmatrix}, \quad (32)$$

where $\mathbf{D}_p$ has full column rank. Then

$$\ker \mathbf{J}_{\text{joint}} = \{0\} \oplus \ker \mathbf{R}_c, \quad \text{rank } \mathbf{J}_{\text{joint}} = d_p + \text{rank } \mathbf{R}_c. \quad (33)$$

The conclusion is independent of the physical endpoint block $\mathbf{R}_p$.

Proof. If $J_{\text{joint}}(\mathbf{x}, \mathbf{y})^\top = 0$, then $D_p \mathbf{x} = 0$. Full column rank gives $\mathbf{x} = 0$, and the remaining equation is $R_c \mathbf{y} = 0$. The kernel and rank formulas follow. $\square$

This is the precise limited sense in which clocks restore selected scale contrasts that path-only null data cannot provide. A globally constant conformal mode, or any conformal mode taking equal values at all sampled endpoints, remains invisible. It is not a full theorem for moving detectors or arbitrary gauges. The operational frame is fixed by static emitter and receiver worldlines, Killing-time normalization, and proper-frequency comparisons. General endpoint bookkeeping belongs in a subsequent paper.

6 Continuum Fisher operator and causal microlocal visibility

A finite ray set gives a distributional rank- m operator. A continuum design measure smooths those ray deltas into the normal operator of Lorentzian integral geometry.

Let $\mathcal{C} = \mathbb{R}^n \times S^{n-1}$ denote the manifold of complete future-directed null rays in the parameterization (3), and equip it with the product measure $d\mu = d\mathbf{y} d\vartheta$, where $d\vartheta$ is standard surface measure. For $\mathbf{u}, \mathbf{v} \in \mathcal{E} = C_c^\infty(\mathbb{R}^{1+n}; S^2 T^* \mathbb{R}^{1+n})$ and a nonnegative angular weight $q \in L^\infty(S^{n-1})$, define

$$F_q(\mathbf{u}, \mathbf{v}) = \int_{S^{n-1}} \int_{\mathbb{R}^n} q(\vartheta) (L_2 \mathbf{u})(\mathbf{y}, \vartheta) (L_2 \mathbf{v})(\mathbf{y}, \vartheta) d\mathbf{y} d\vartheta. \quad (34)$$

Compact support makes this a continuous nonnegative bilinear form on $\mathcal{E} \times \mathcal{E}$. For the delay convention $A = L_2/2$, the factor $4 \text{Var}(\hat{\nu})$ cancels the two factors of $1/2$, so q may be read directly as photon density times frequency variance. A bounded weight depending also on $\mathbf{y}$ still defines $L_2^* Q L_2$, but generally destroys translation invariance and the Fourier-multiplier formula below.

Theorem 6.1 (*Continuum Fisher operator*). *Let $n \geq 3$, let $\mathcal{E} = C_c^\infty(\mathbb{R}^{1+n}; S^2 T^* \mathbb{R}^{1+n})$, and let $q \in L^\infty(S^{n-1})$ be nonnegative. The form (34) defines a continuous operator $\mathcal{N}_q^Q : \mathcal{E} \rightarrow \mathcal{E}'$ by*

$$\mathcal{N}_q^Q = L_2^* Q L_2, \quad (35)$$

where Q is multiplication by q . With the Fourier convention $\widehat{H}(\xi) = \int_{\mathbb{R}^{1+n}} e^{-ix \cdot \xi} H(x) dx$, $H(x) = (2\pi)^{-(n+1)} \int e^{ix \cdot \xi} \widehat{H}(\xi) d\xi$, and the product ray measure above, the translation-invariant Fourier multiplier is

$$\sigma(\mathcal{N}_q^Q)(\xi)H = 2\pi \int_{S^{n-1}} q(\vartheta) \delta(\xi_0 + \vartheta \cdot \xi') \langle H, k_\vartheta^{\otimes 2} \rangle k_\vartheta^{\otimes 2} d\vartheta. \quad (36)$$

Here $\langle \cdot, \cdot \rangle$ is the componentwise Euclidean pairing on symmetric tensors in the chosen Minkowski coordinates. Consequently,

$$\langle H, \sigma(\mathcal{N}_q^Q)(\xi)H \rangle = 2\pi \int_{S^{n-1}} q(\vartheta) \delta(\xi_0 + \vartheta \cdot \xi') |H_{\mu\nu} k_\vartheta^\mu k_\vartheta^\nu|^2 d\vartheta \geq 0. \quad (37)$$

Proof. By definition of the distributional adjoint, for $u, v \in \mathcal{E}$, $\langle L_2^* Q L_2 u, v \rangle = \langle Q L_2 u, L_2 v \rangle_{L^2(\mathcal{C}, d\mu)} = F_q(u, v)$, which proves (35) and continuity into $\mathcal{E}'$. Insert the displayed Fourier representation of u into the line integral. Integration in the affine parameter gives $2\pi\delta(\xi_0 + \vartheta \cdot \xi')$, while integration over the full intercept $y \in \mathbb{R}^n$ enforces equality of the spatial Fourier variables in the adjoint. The two tensor contractions contribute $k_\vartheta^{\otimes 2}$ on the input and output fibers, yielding (36); pairing with H gives (37). Appendix B records the same calculation explicitly. $\square$

The delta constraint gives an immediate causal stratification.

6.1 Timelike covectors: microlocally invisible

If $|\xi_0| > |\xi'|$, then no $\vartheta \in S^{n-1}$ satisfies $\xi_0 + \vartheta \cdot \xi' = 0$. Hence $\sigma(\mathcal{N}_q^Q)(\xi) = 0$. Metric oscillations whose wavefront covectors are timelike have spacelike phase fronts. Restriction to every null ray is nonstationary, so the oscillations average out. In statistical language, the local QFI density vanishes in these directions. The statement is microlocal: the principal Fisher symbol annihilates timelike covectors. It does not assert that compactly supported information in those directions could never be inferred by unstable analytic continuation under additional hypotheses.

6.2 Spacelike covectors: visible modulo gauge

If $|\xi_0| < |\xi'|$, the constraint set is a nonempty $(n-2)$ -sphere. The delta scaling produces an order -1 symbol. Potential and conformal tensors remain in its fiberwise radical. Here $\xi \odot v := \xi \otimes v + v \otimes \xi$. If $H = \xi \odot v$, then on the constraint

surface $H(\mathbf{k}, \mathbf{k}) = 2\xi(\mathbf{k})v(\mathbf{k}) = 0$, and if $H = \phi\eta$, then $H(\mathbf{k}, \mathbf{k}) = 0$ because $\mathbf{k}$ is null. Under complete angular coverage and the hypotheses of the cited tensor-transform results, an appropriate gauge projection is elliptic on the spacelike cone [2, 5, 7].

6.3 Lightlike covectors: the transition

If $|\xi_0| = |\xi'|$, the angular level set degenerates. The full normal operator is not globally a classical pseudodifferential operator; its Schwartz kernel belongs to a paired-Lagrangian class, and the light cone is the transition between the diagonal and flow-out relations [2, 3]. This caveat should be promoted rather than hidden:

Quantum-statistical identifiability is stratified by the causal type of the covector.

On-shell gravitational waves lie exactly on this delicate lightlike boundary. Finite-link and endpoint-sensitive detectors can retain nonzero response there, but that response is not governed by the ordinary spacelike ellipticity argument developed here.

6.4 Visibility spectra require normalization

Eigenvalues of a QFIM depend on parameter units. Let $\mathbf{R} \succ \mathbf{0}$ be a declared parameter metric, prior precision, Sobolev penalty, or energy norm. The declared finite-dimensional object is

$$\tilde{\mathbf{F}} = \mathbf{R}^{-1/2} \mathbf{F} \mathbf{R}^{-1/2}. \quad (39)$$

Microlocally, an order-compensated visibility form may be written schematically as

$$\mathcal{V}(\mathbf{x}, \xi) = |\xi| \Pi_\xi \sigma^{-1}(\mathcal{N}_q^Q)(\mathbf{x}, \xi) \Pi_\xi, \quad (40)$$

where Π_ξ projects onto a chosen gauge-fixed or quotient fiber. The generalized eigenvalues of the pair $(\mathbf{F}, \mathbf{R})$ are coordinate-consistent once $\mathbf{R}$ is declared, and (39) is one representation of that generalized problem; the particular diagonal $\mathbf{R} = \text{diag}(|J_{\cdot j}|^2)$ used in the experiments below is a declared column-energy normalization and a modeling choice, not an invariant of the physics. Small generalized eigenvalues then have a literal statistical interpretation: they are directions in which unbiased local estimation has large variance and the inverse problem is unstable.

7 Finite ray networks and optimal design

Let $V \subset \mathcal{Q}$ be a d -dimensional gauge-quotiented model family with basis $e_1, \dots, e_d$ and parameter vector θ . For candidate ray γ_a , define $J_{ai} = A_{\gamma_a} e_i$. For independent Gaussian packets,

$$F = J^\top W J, \quad W = \text{diag}(s_1^{-2}, \dots, s_m^{-2}). \quad (42)$$

Theorem 7.1 (*Finite identifiability and minimum scalar channels (Theorem B)*). Let $a(\gamma) \in \mathbb{R}^d$ denote the sensitivity vector of a scalar null channel after normalization by a fixed $R \succ 0$.

1. A selected network Γ is locally identifiable on V if and only if $\text{span}\{a(\gamma) : \gamma \in \Gamma\} = \mathbb{R}^d$. Equivalently, $\tilde{F} \succ 0$.
2. Fewer than d scalar channels cannot identify a d -dimensional family.
3. If the full candidate family spans $\mathbb{R}^d$, then some set of exactly d candidate rays is algebraically identifying.

Proof. The R -normalized Fisher matrix is a positive weighted sum of outer products $a(\gamma)a(\gamma)^\top$. Its kernel is the orthogonal complement of the span of the selected feature vectors. Parts (i) and (ii) follow. If the full family spans $\mathbb{R}^d$, elementary linear algebra permits selection of a basis of d feature vectors, proving (iii). $\square$

The theorem concerns rank, not stability. A nearly dependent d -ray basis may be unusable under noise. Design should therefore control the spectrum of (39).

For a discrete candidate pool and nonnegative photon allocations w_a , write $\tilde{F}(w) = \sum_{a=1}^M w_a \tilde{a}_a \tilde{a}_a^\top$. Standard criteria include

$$\text{E-optimal:} \quad \max_w \lambda_{\min}(\tilde{F}(w)) \quad (44)$$

$$\text{D-optimal:} \quad \max_w \log \det \tilde{F}(w) \quad (45)$$

$$\text{A-optimal:} \quad \min_w \text{tr}(\tilde{F}(w)^{-1}) \quad (46)$$

The continuous-weight versions are classical convex or concave design problems under a linear budget [21]. Selecting exactly K distinct rays is combinatorial. The experiments below compare continuous relaxation plus swap refinement with a rank-one-updated greedy D-optimal rule. Over the registered regime,

the deterministic greedy method is already millisecond scale after $\mathcal{J}$ has been assembled; amortization is therefore useful only if future tasks enlarge the candidate pool, make the forward model expensive, introduce correlated-probe co-design, or impose stricter online constraints.

8 One unrepeated photon: binary geometry discrimination

Continuous-parameter QFI is an asymptotic local object. If “single photon” is meant literally as one unrepeated use, binary discrimination is the operational question. Let two candidate weak geometries h_0 and h_1 induce delays $\tau_i = A_\gamma h_i$, $\Delta\tau = A_\gamma(h_1 - h_0)$.

Proposition 8.1 (*Gaussian Helstrom bound*). *For equal prior probabilities and the Gaussian temporal mode (9), the minimum one-shot error is*

$$P_{\text{err}}^* = \frac{1}{2} \left(1 - \sqrt{1 - \exp\left(-\frac{[A_\gamma(h_1 - h_0)]^2}{4s^2}\right)} \right). \quad (47)$$

If $h_1 - h_0 \in \ker A_\gamma$, then $P_{\text{err}}^* = 1/2$ and no quantum measurement beats guessing. For small delay differences,

$$P_{\text{err}}^* = \frac{1}{2} - \frac{|A_\gamma(h_1 - h_0)|}{4s} + O\left(\frac{|A_\gamma(h_1 - h_0)|^3}{s^3}\right). \quad (48)$$

Proof. The squared overlap of the shifted Gaussian states is

$|\langle \psi_{\tau_0} | \psi_{\tau_1} \rangle|^2 = \exp[-(\Delta\tau)^2/(4s^2)]$. The equal-prior Helstrom formula for two pure states gives (47) [22]; Taylor expansion gives (48). $\square$

9 Computational verification and finite-network experiments

Computational methods. Every code path used for a reported result was reviewed by the authors, executed under the pinned stop-on-failure checks, and reconciled against the committed machine-readable reports. Generative-AI assistance is disclosed once, in the Acknowledgments; model names, versions, and session-level logs are recorded in the repository.

The released package separates proof, numerical implementation checks, finite-design measurements, and exploratory ML. All reported geometric Jacobians in the final campaign use Gauss–Legendre order 1024. The numerical controls used in this revision correspond to source snapshot a384312; the analysis-only candidate-density and ray-budget reduction corresponds to fdd6cfa. The full hashes and chronology are recorded in `REGISTRATION.md`.

9.1 Implementation checks

Seven scientific or numerical checks and one process control validate independent code paths for the proved identities. The checks do not constitute additional proofs.

Table 1: Implementation checks and observed values. Gate 7 is a process control rather than a measurement.

No.	Check	Observed result
1	Componentwise conformal contraction	$A_{\Gamma}(\phi\eta)$ relative residual $2.61 \times 10^{-17} < 10^{-14}$
2	Compact-support potential mode	$A_{\Gamma}(2d_s V)$ $4.07 \times 10^{-14} < 10^{-10}$, with order-convergence report
3	Gauge persistence over all 288 rays	maximum residual $1.21 \times 10^{-12} < 10^{-10}$
4	$\text{rank } F = \text{rank } J$ under matched SVD/eigenvalue thresholds	passed for every tested ray prefix
5	Gaussian packet QFI scaling	slope -2.0000 ; variance relative error 6.66×10^{-16}
6	Non-polynomial endpoint identity	relative error $1.51 \times 10^{-14} < 10^{-8}$
7	Stop rather than patch any failure	writes the report and raises <code>SystemExit</code>
8	Physical-family independence modulo sampled gauge	margins $1.92 \times 10^6, 7.73 \times 10^5, 7.46 \times 10^5, 6.86 \times 10^5$ for $d = 6, 8, 12, 16$

Gate 8 compares the smallest physical singular value with a sampled potential-plus-conformal numerical floor. The large margins certify the finite families used in the campaign, not an infinite-dimensional quotient theorem.

9.2 Standalone static-redshift verification

The static experiment uses four localized stationary conformal modes, five clocks, ten oriented clock links, and 144 delay rays. The componentwise delay block has maximum magnitude 3.41×10^{-18} and numerical rank zero. Two independent implementations of $R_{AB}(\phi\eta) = \frac{1}{2}[\phi(A) - \phi(B)]$ agree exactly, and the endpoint matrix has rank four. Four greedy-selected links realize rank four, while three links realize rank three. A nonlinear lapse check based on $N = \sqrt{1 + \varepsilon\phi}$ converges to the linear functional with observed second-order slope 2.00017.

For stationary gauge modes with $V_0 = 0$ and fixed clock endpoints, the assembled endpoint tensor is nonzero (maximum norm 8.56×10^{-2}) while the redshift response is zero. The same modes were integrated along the delay rays: the residual falls from 1.37×10^{-8} at order 1024 to 2.05×10^{-11} at order 2048 and 4.40×10^{-15} at order 4096. The joint experiments therefore use order 2048 for this narrow-bump gauge stress test.

9.3 Experiment D+R-1: a distinct observable lifts the kernel

The joint family is

$$h_{\theta,\alpha} = \sum_{i=1}^{d_p} \theta_i e_i + \sum_{j=1}^4 \alpha_j \phi_j \eta, \quad d_p \in \{6, 8, 12, 16\}.$$

All physical modes are stationary and satisfy $h_{0i} = 0$, so the endpoint formula applies. Scalar physical modes have nonzero R_p , making Theorem 5.3 nontrivial rather than block diagonal. For every d_p , the observed ranks are identical in form:

Arm	Added resource	observed rank / nullity
A	selected delay rays only	$d_p / 4$
B	same rays plus 3 clock links	$d_p + 3 / 1$
C	same rays plus 4 clock links	$d_p + 4 / 0$
D	same rays plus 4 further delay rays	$d_p / 4$
E	same rays plus all 10 clock links	$d_p + 4 / 0$
F	arm E plus a constant conformal mode	$d_p + 4 / 1$

The complete 144-ray delay bank also has rank d_p and conformal nullity four. Thus the negative-control conclusion is independent of how the additional delay rays are selected. The constant conformal mode localizes the arm-F kernel vector with weight one, as expected because all clock endpoints see the same value.

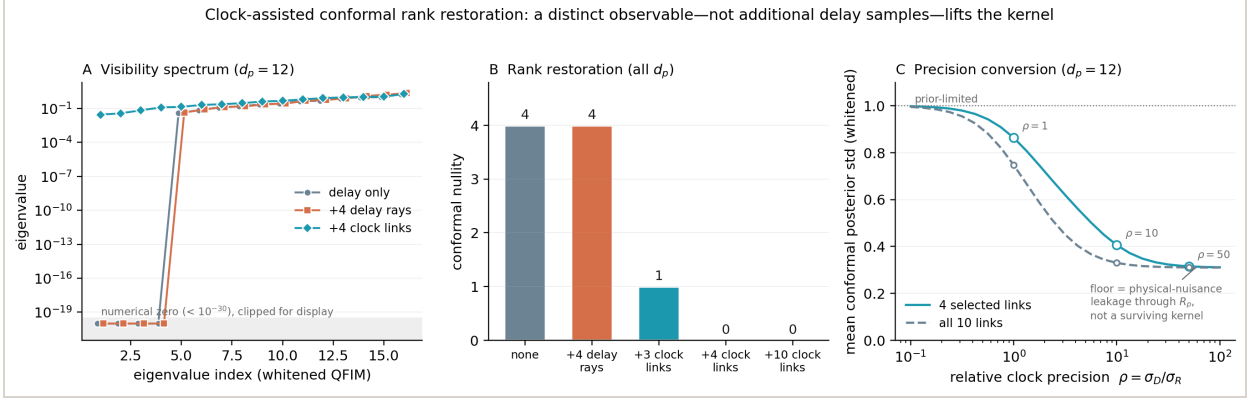


Figure 3: Clock-assisted conformal rank restoration. (A) Ordered eigenvalues of the R -normalized QFIM for $d_p = 12$: delay-only sensing and four additional delay observations retain four numerical-zero conformal directions, whereas four selected static-clock comparisons make the complete spectrum positive. (B) Conformal nullity, common to $d_p \in \{6, 8, 12, 16\}$, remains four after additional delay observations, falls to one with three independent clock links, and vanishes with four. (C) Mean conformal posterior standard deviation in R -normalized coordinates versus relative clock precision $\rho = \sigma_D/\sigma_R$; the nonzero high- ρ floor results from finite-delay uncertainty in physical nuisance modes entering R_p , not residual conformal non-identifiability. Values are drawn from the committed report.

For $d_p = 12$, four clock links raise the smallest eigenvalue of the R -normalized QFIM from numerical zero (below 10^{-34}) to 2.81×10^{-2} . Identifiability and precision are deliberately separated. In raw mode coordinates with unit channel noise and an identity prior, mean conformal posterior standard deviation changes only from 1 to approximately 0.9995, because the synthetic endpoint amplitudes are small. In R -normalized coordinates it is 0.863 at relative precision $\rho = \sigma_D/\sigma_R = 1$ and decreases as clock precision improves. Its nonzero high- ρ floor is not a surviving conformal kernel: finite-delay uncertainty in physical modes propagates through the nonzero nuisance block R_p .

9.4 Registered deterministic full-metric campaign

The registered surface is $M \in \{256, 512, 1024, 4096\}$, $d \in \{6, 8, 12, 16\}$, $K/d \in \{1, 1.25, 1.5, 2\}$, with seeds $\{2026, 3407, 9181, 17041, 27183\}$. For the 320-record greedy campaign, each order-1024 geometric Jacobian was assembled once per candidate-pool size and reused across dimensions, budgets, and packet-width seeds. Packet widths are log-uniform on $[0.05, 0.20]$ and $q_a = s_a^{-2}$. A pinned contiguous 20% angular sector is suppressed and the design is reoptimized to measure robustness. Posterior RMSE and worst-direction variance are analytic; 95% interval coverage is checked by Monte Carlo only on three registered representative cells.

All 320 records are full rank. Across the complete surface, the R -normalized minimum eigenvalue ranges from 0.083 to 4.27, and the median blocked-to-unblocked ratio is 0.989. The per-cell median ratio ranges from 0.63 to 1.02; values slightly above one occur because the blocked and unblocked designs are separately optimized by a non-global greedy heuristic. Coverage on the representative cells is 0.945–0.948, as expected in the correctly specified Gaussian model.

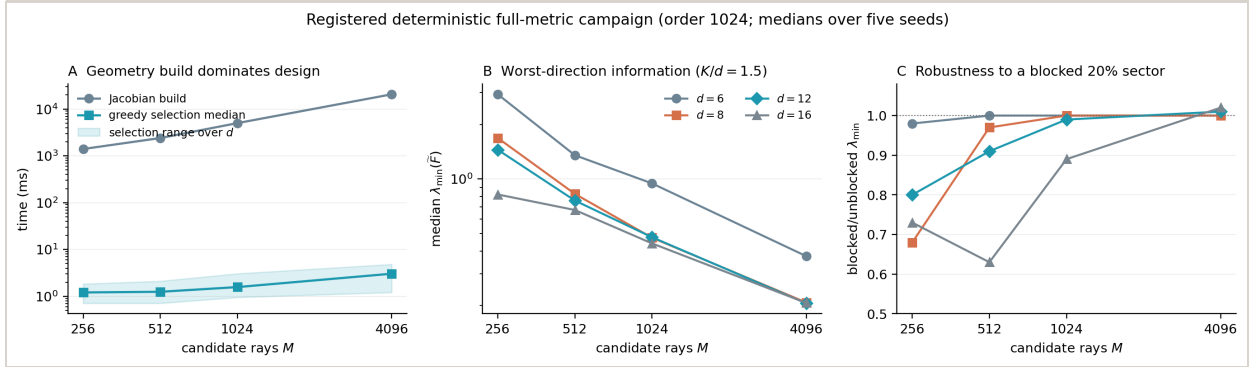


Figure 4: Summary of the repository-registered deterministic campaign at $K/d = 1.5$. (A) Jacobian assembly dominates selection by several orders of magnitude across the candidate pool. (B) Worst-direction information declines with pool size under the M -dependent R -normalization, so cross- M changes reflect the declared metric and fixed selected fraction, not raw physical information monotonicity. (C) Blocking ratios compare separately reoptimized greedy designs under a pinned 20% angular-sector exclusion. Every plotted value is drawn from the committed report and reproduced in Table 6.

Per-cell medians for all 16 design cells at $K/d = 1.5$ are listed in Appendix D, Table 6.

At $K/d = 1.5$, median R -normalized posterior RMSE ranges from 0.263 to 0.627, while raw-coordinate RMSE ranges from 0.895 to 0.944. This contrast illustrates why both coordinate conventions are reported. Greedy selection takes roughly 0.7–4.8 ms in the compact table, whereas Jacobian assembly takes 1.4–20.6 s. A high-quality relaxed-E-plus-swap reference on three representative cells improves $\lambda_{\min}$ by factors 2.05, 2.96, and 3.67 over greedy D-optimal selection, but costs 1.1, 5.7, and 41.3 s per instance, respectively. Greedy D-optimality is therefore the deployment-scale method, while relaxed-E quantifies the price of optimizing a different spectral objective.

Candidate density and ray budget. The registered surface already contains the density and budget ablations, but cross-pool interpretation requires care. The full-bank metric $R = \text{diag}(\|J_{:,j}\|^2)$ is rebuilt for each (M, d) and grows approximately linearly with the number of candidate rays. Raw R -normalized $\lambda_{\min}$ values are therefore not comparable across M . The post-registration reduction in Table 3 uses two cross- M quantities instead: raw-coordinate posterior RMSE under the same unit-noise, standard-normal convention, and the declared pool-size adjustment $M\lambda_{\min}(\tilde{F})$. Along the budget axis at fixed (M, d) , the normalization is identical and ordinary R -normalized comparisons are valid.

Table 2: Candidate-density and ray-budget reduction of the 320-record campaign. Cross- M spectral values are not compared directly because the full-bank normalization matrix $\mathbf{R}$ changes with M ; Panel A therefore reports raw-coordinate posterior RMSE and the declared pool-size-adjusted quantity $M\lambda_{\min}(\tilde{\mathbf{F}})$. Panel B compares budgets at fixed $M = 1024$, where the normalization is identical. Entries are medians over the five registered seeds.

A. Candidate density at fixed $K/d = 1.5$: $M = 256 \rightarrow 4096$

d	raw RMSE 256 $\rightarrow$ 4096	$M\lambda_{\min}$ gain	latency factor	final selection (ms)
6	0.9220 $\rightarrow$ 0.8949	2.05×	1.7×	1.21
8	0.9296 $\rightarrow$ 0.8979	1.98×	1.9×	1.76
12	0.9421 $\rightarrow$ 0.9192	2.28×	3.2×	4.83
16	0.9440 $\rightarrow$ 0.9206	4.03×	2.3×	4.32

B. Ray budget at fixed $M = 1024$: $K/d = 1 \rightarrow 2$

d	$\lambda_{\min}$ gain	RMSE reduction	condition-number reduction	last-step gain 1.5 $\rightarrow$ 2
6	3.19×	1.27× lower	2.0× lower	1.01×
8	2.39×	1.25× lower	1.2× lower	1.33×
12	2.47×	1.27× lower	1.3× lower	1.39×
16	2.82×	1.29× lower	1.6× lower	1.25×

Increasing the pool 16-fold from $M = 256$ to 4096 at fixed $K/d = 1.5$ improves the pool-adjusted worst-direction statistic by factors 1.98–4.03, with the largest gain at $d = 16$, while median selection remains below 5 ms. Raw-coordinate RMSE improves only modestly because the declared unit-noise prior dominates the absolute scale. At fixed $M = 1024$, doubling the budget from $K/d = 1$ to 2 improves $\lambda_{\min}$ by factors 2.39–3.19 and lowers $\mathbf{R}$ -normalized posterior RMSE by factors 1.25–1.29. The final step from $K/d = 1.5$ to 2 is saturated at $d = 6$ (1.01×) but remains informative at $d \geq 8$ (1.25–1.39×). Every one of the 320 records, and every separately redesigned blocked counterpart, remains full rank; the worst individual blocked-sector retention is 0.188. On this surface, candidate density and ray budget price conditioning and precision rather than identifiability.

9.5 Submission controls: packet weights, gauge quotient, and design objective

Three additional controls address the most direct alternative explanations.

QFI weighting and packet heterogeneity (W-1). A registered 2×2 control crossed selection weights (uniform versus $q_a = s_a^{-2}$) with packet resources (information-matched homogeneous versus heterogeneous widths) on the three representative cells and five seeds. Every design was cross-evaluated under the true objective of its resource condition, and the weights were normalized so that $M^{-1} \sum_a q_a = 1$ in every arm. The median results are summarized in Table 3.

M	d	K	QFI-aware / blind $\lambda_{\min}$	set overlap	heterogeneous / homogeneous $\lambda_{\min}$
256	12	18	2.36	0.28	1.94
1024	12	18	2.32	0.11	2.08
4096	16	24	7.18	0.04	2.82

Table 3: Packet-weight control W-1. Entries are medians over the five registered seeds. All arms satisfy $M^{-1} \sum_a q_a = 1$; homogeneous packets are information matched to the heterogeneous draw. The fourth column cross-evaluates QFI-aware and weight-blind designs under the true heterogeneous QFI objective.

The QFI-aware design improves the true weighted $\lambda_{\min}$ by factors 2.36, 2.32, and 7.18 over weight-blind selection, while selected-set overlap falls from 28% to 4% as the pool grows. Thus packet bandwidth changes which rays should be measured, not merely the numerical scale of an unchanged design. Under QFI-aware selection, heterogeneous resources improve $\lambda_{\min}$ by factors 1.94–2.82 over information-matched homogeneous resources. That comparison must be read with its normalization in view. Here q_a is an exogenous candidate quality, and the weights are matched over the whole candidate pool rather than over the selected subset, so a QFI-aware selector may carry a larger selected resource $\sum_{a \in S} q_a$ in the heterogeneous arm. The supported claim is therefore that heterogeneous link quality changes which rays should be selected; it is not a claim that an experimenter who reallocates a fixed photon-number, bandwidth, or energy budget across a fixed selected set gains that factor. No directional success threshold was specified for this control; the measured differences are the result.

No-quotient negative control (NQ-1). An unquotiented family with 12 physical, two conformal, and two potential-gauge columns remains rank 12, hence nullity four, for ray budgets 16, 32, 64, 128, and 288. The numerical scales

are separated explicitly:

$$\max |J_{\text{conf}}| = 1.2 \times 10^{-18}, \quad \|J_{\text{pot}}\|_{\text{floor}} = 2.4 \times 10^{-10}, \quad \sigma_{\min}(J_{\text{phys}}) = 6.9 \times 10^{-2}.$$

The rank threshold lies between the quadrature gauge floor and the physical spectrum, so numerical gauge leakage is never counted as recovered information. The quotient is therefore operationally necessary, not merely notational.

D-optimal versus E-optimal design. The representative relaxed-E runs were cross-scored on both objectives and on posterior metrics; Table 4 prevents the circular comparison in which each design is judged only by the objective it optimized.

M	d	K	design	$\lambda_{\min}$	log det	RMSE	worst σ	time
256	12	18	greedy-D	1.650	38.6	0.263	0.614	1 ms
256	12	18	relaxed-E	3.382	29.9	0.324	0.478	1.1 s
1024	12	18	greedy-D	0.477	23.6	0.406	0.823	1 ms
1024	12	18	relaxed-E	1.413	17.2	0.466	0.644	5.7 s
4096	16	24	greedy-D	0.175	16.6	0.545	0.923	5 ms
4096	16	24	relaxed-E	0.642	8.8	0.619	0.780	40.5 s

Table 4: Cross-objective comparison at the three representative cells, seed 2026, order 1024, in $\mathbf{R}$ -normalized coordinates. Each selected design is scored on both E- and D-optimal criteria and on posterior metrics.

Relaxed-E improves the minimum eigenvalue and worst-direction posterior error; greedy-D improves log-determinant and average RMSE at three to four orders of magnitude lower latency. Neither method dominates independently of the scientific loss function. Candidate density and K/d are reduced explicitly in Table 2, and the consolidated numerical-order checks are reported in Appendix C.

9.6 One-shot amortized selection: a negative result

A permutation-equivariant DeepSets policy [24] was trained to score candidate rays and return one hard top-K set. On 100 held-out tasks it clears the two registered weak baselines, with median paired $\lambda_{\min}$ ratios of 3.57× over uniform random and 1.92× over angular spread and 100% full-rank success, but it

reaches only median $\lambda_{\min} = 0.1202$ against 2.1746 for greedy D-optimal selection at roughly 45× lower latency. Its median operating point is therefore Pareto dominated in both objective value and latency. The full held-out evaluation, the leverage-score control, and the structural diagnosis are given in Appendix E.

9.7 Completed and outstanding computational scope

The delay-versus-redshift, QFI-weighting, packet-homogeneity, no-gauge-quotient, D-versus-E, candidate-density, and ray-budget analyses are complete. Quadrature stability is consolidated in Appendix C. The sole registered scientific ablation deferred beyond Paper 1 is the independent-versus-correlated probe comparison with off-diagonal W_ρ , which belongs to the quantum-network companion paper. The PTA/LISA gate remains unrun and is not used to support the spacelike microlocal result.

10 Discussion

10.1 What is mathematically new

The algebraic pullback $J^T W J$ is elementary. Its placement on the Lorentzian symmetric-tensor light-ray transform is what gives it content. It derives the rank of a photon probe from a geometric forward map, turns the transform kernel into a quantum-statistical radical, distinguishes universal gauge blindness from finite experimental blindness, and identifies the continuum Fisher operator with a microlocally stratified Lorentzian normal operator. The closest statistical precedent is the Riemannian scalar Fisher operator $I^* I$ [8]; the closest quantum precedents estimate selected metric parameters or intervals [10, 11, 13]. The present framework adds the tensor gauge quotient, causal microlocal symbol, finite-network rank law, and explicit quantum non-lifting statement.

10.2 A new observable versus more samples

The block-rank proposition and D+R-1 isolate a general experimental principle. The complete delay bank has zero conformal columns, so no choice or number of delay rows can lift that nullspace. Clock comparisons alter the forward

operator and add R_c . This is stronger than showing that one particular finite delay selection is deficient: the full candidate bank retains the same four-dimensional conformal kernel.

The same logic applies to quantum probe design. For any $W_\rho \succeq 0$, $\ker J \subseteq \ker(J^\top W_\rho J)$. Entanglement, squeezing, or collective readout can improve a visible mode, suppress a nuisance combination, or redistribute the visibility spectrum. They cannot reveal a direction absent from the physical channel. Deflection, polarization transport, massive probes, or clock-redshift links are valuable precisely because they change J .

10.3 Identifiability, conditioning, and precision

Rank restoration is noise independent; useful posterior contraction is not. The raw-coordinate clock rows in the synthetic experiment are small relative to the declared unit noise, so full rank coexists with weak raw posterior contraction. Normalizing by the full candidate-bank column energy removes arbitrary diagonal basis scaling and exposes a stronger but still finite precision gain. Neither coordinate convention is universally physical. A detector-facing analysis must replace synthetic normalization with actual timing, frequency, photon-number, and prior units.

The high-clock-precision floor in Fig. 3 is also informative. The endpoint equations mix physical and conformal coefficients through R_p . When clock noise vanishes but delay noise remains finite, uncertainty in physical nuisance modes propagates into the conformal estimate. Full rank guarantees a positive Fisher eigenvalue; it does not imply zero posterior variance at finite resources.

10.4 Deterministic design versus amortization

The registered campaign shows that geometric Jacobian construction, not subset selection, is the dominant cost in the studied regime. The density reduction reinforces this separation: increasing M from 256 to 4096 improves the pool-adjusted worst-direction statistic by up to 4.03×, yet greedy selection remains below 5 ms while Jacobian assembly is second scale. A learned policy that consumes J -derived features cannot amortize the expensive stage; it can only replace a millisecond greedy step. The negative DeepSets result should therefore not be read as a general failure of learned experimental design. It

narrows the activation conditions: amortization becomes relevant for much larger candidate pools, expensive state-dependent forward models, strict online redesign, or joint discrete-continuous quantum-network resources.

10.5 Limitations

Several limitations are substantive.

First, the leading delay is achromatic only in geometric optics. Dispersion, finite aperture, curvature-induced mode mixing, scattering, and loss produce a more general channel; the pullback principle survives, but both $\mathbf{J}$ and $\mathbf{W}_\rho$ can become state dependent.

Second, full endpoint gauge invariance for moving emitters and receivers requires varying worldlines, clock synchronization, tetrads, and source phases together with the metric. The static section is intentionally restricted to stationary $\mathbf{h}$ with $\mathbf{h}_{0i} = \mathbf{0}$ and a fixed operational clock frame.

Third, continuum ellipticity holds only microlocally on the spacelike cone after gauge treatment and adequate angular coverage. Timelike directions are microlocally invisible to the normal operator; the light cone requires paired-Lagrangian analysis. PTA and interferometric responses lie on this delicate on-shell boundary.

Fourth, the finite families use localized synthetic modes. These are kinematic test families: they are not constrained to satisfy the linearized Einstein equations or a specified source model, so on-shell field equations may remove, couple, or reparameterize some of the directions used here. Large quotient margins and cross-seed stability establish numerical independence for those families, not universal physical completeness.

Fifth, the no-gauge, channel-weight, packet-homogeneity, and D-versus-E controls are now complete, but they remain finite synthetic tests rather than detector calibrations.

Sixth, D+R-1 models delay and redshift noise with independent blocks. In an actual implementation, the delay observable itself uses clocks or phase references, so common oscillator, synchronization, and reference noise

can correlate the two blocks. The appropriate model is the joint $\mathbf{J}_{DR}^* \mathbf{W}_\rho \mathbf{J}_{DR}$ form above, with off-diagonal QFI or covariance terms; this is the natural entry point for the quantum-network and error-correction companion study.

Seventh, the information object throughout is the SLD quantum Fisher matrix. For general correlated multiparameter probes the SLD bound is not automatically simultaneously attainable, and the operational benchmark is a compatibility or Holevo-bound analysis [23]. The executed campaigns use independent Gaussian shift channels, for which product time-of-arrival measurement attains the stated QFI, so the non-lifting result is unaffected; statements interpreting an arbitrary $\mathbf{J}^T \mathbf{W}_\rho \mathbf{J}$ as directly achievable precision are not.

10.6 Nonlinear and detector-facing extensions

A natural second paper replaces the abstract family by invariantly parameterized Schwarzschild and Kerr models. A Schwarzschild study should jointly treat areal emitter and receiver radii, impact parameter, delay, redshift, and deflection, then compute nuisance kernels and design spectra. Existing work already treats selected photon-QFI parameters [11, 12]; the differentiated contribution is joint identifiability and resource design.

Kerr adds polarization transport and gravitational Faraday rotation. A rank-jump theorem requires a configuration in which scalar observables remain degenerate in a spin direction while polarization adds an independent derivative; spin sensitivity cannot be assumed absent from delay or deflection. A separate quantum-network paper should implement general off-diagonal $\mathbf{W}_\rho$, product/GHZ/graph probes, noise-aware covariance design, and quantum-error-corrected nuisance rejection. PTA and LISA/TDI should remain an application paper until deterministic link responses, endpoint terms, stochastic averaging, and the Hellings–Downs curve are reproduced quantitatively.

11 Conclusion

Weak null propagation supplies a canonical map from metric perturbations to quantum-channel parameters. For a one-photon temporal packet, the first-order delay is the symmetric-tensor light-ray transform and its QFI is

rank one. Finite networks pull back the probe QFI to $\mathbf{J}^\top \mathbf{W}_\rho \mathbf{J}$; continuum networks produce the weighted Lorentzian light-ray normal operator. The resulting visibility is stratified by causal covector type and obstructed by potential, conformal, and incomplete-sampling kernels.

The quantum non-lifting principle is exact: a probe state can reshape information only within the image of the physical forward map. The static clock channel demonstrates the complementary possibility. Delay rows annihilate conformal perturbations, while endpoint redshift rows restore exactly the rank carried by $\mathbf{R}_c$. In four tested kinematic families, three clock links raised the joint rank by three, four raised it by four, and the complete 144-ray delay bank left the conformal nullity unchanged. The experiment therefore realizes the statement that a distinct observable, not additional samples of an annihilating channel, lifts the kernel.

The registered deterministic campaign shows that these finite quotient models can be designed robustly: all 320 records — 64 design cells crossed with five packet-width seeds — are full rank, the median blocked-sector information ratio is 0.989, and subset selection is millisecond scale once the geometric Jacobian is known. The campaign reduction shows that the remaining finite resources price conditioning rather than identifiability: a 16-fold denser candidate pool raises the declared pool-adjusted worst-direction statistic by up to 4.03 $\times$, while doubling $\mathbf{K}/d$ raises the fixed-pool $\mathbf{R}$ -normalized minimum eigenvalue by 2.39–3.19 $\times$. A one-shot DeepSets selector provides a useful negative result: it improves over weak baselines, but its median operating point is dominated by deterministic Fisher-aware selection. The remaining work is not to establish the central phenomenon, but to integrate it into detector-scale models and optimize correlated quantum probe covariance on the already identified geometric image.

A single photon does not define a spacetime. It witnesses one rank-one functional of it. Networks of photons and clocks build a gauge-reduced statistical geometry whose visible directions are determined first by physics, then by resources.

Acknowledgments and AI-assistance disclosure

The numerical work relied on open-source scientific-computing and typesetting tools. Generative-AI systems were used as research-assistance tools for literature organization, candidate derivations, code drafting and debugging, adversarial checks, manuscript revision, and figure scripting. The authors set the research questions, model scope, experiment registrations, and acceptance gates; independently reviewed all analytic derivations, numerical outputs, and citations; executed or supervised the numerical runs; and edited every scientific claim. The authors accept full responsibility for the manuscript. No AI system is an author, and no generative model was used to fabricate observational data or substitute for the stated numerical calculations. Model names, versions, and session-level logs are recorded in the repository.

A Distributional finite-ray adjoint

For $h \in \mathcal{E}$ and $c \in \mathbb{R}^m$, $\langle A_\Gamma h, c \rangle = \frac{1}{2} \sum_{a=1}^m c_a \int h_{\mu\nu}(\gamma_a(\lambda)) k_a^\mu k_a^\nu d\lambda$. Define the tensor distribution $k_a^{\otimes 2} \delta_{\gamma_a}$ by $\langle k_a^{\otimes 2} \delta_{\gamma_a}, h \rangle = \int h_{\mu\nu}(\gamma_a(\lambda)) k_a^\mu k_a^\nu d\lambda$. Then (16) follows. This is continuous in the standard test-function topology but not as a bounded functional on an unweighted spacetime L^2 space.

B Fourier representation of the continuum symbol

For a translation-invariant ray family, substitute the Fourier representation $h_{\mu\nu}(x) = (2\pi)^{-(n+1)} \int e^{ix \cdot \xi} \widehat{h}_{\mu\nu}(\xi) d\xi$ into (4). Integration in λ gives $\int_{\mathbb{R}} e^{i\lambda(\xi_0 + \vartheta \cdot \xi')} d\lambda = 2\pi \delta(\xi_0 + \vartheta \cdot \xi')$. Integration over the full intercept $y \in \mathbb{R}^n$ imposes the matching spatial Fourier variable in the adjoint. With standard surface measure in ϑ , the remaining angular integral yields (36) with coefficient 2π . Homogeneity of the delta distribution contributes order -1 on the nondegenerate spacelike cone.

C Numerical convergence summary

Check	Measured evidence
Compact-gauge cancellation	6.04×10^{-8} , 2.90×10^{-8} , 1.05×10^{-10} , and 4.07×10^{-14} at orders 128, 256, 512, 1024 on the single-ray check; all-ray prefix maximum 1.21×10^{-12} at order 1024.
Gaussian packet variance	Quadrature agrees with $1/(4s^2)$ to relative error 6.7×10^{-16} over the 17-point width grid; fitted slope -2.0000000000 .
Endpoint gauge identity	Non-polynomial-profile relative error 1.51×10^{-14} at order 1024.
Design-order stability	Boundary-cell selected subsets are identical between orders 512 and 1024 for all five seeds; maximum relative $\lambda_{\min}$ shift 2.9×10^{-8} .
Narrow static-gauge modes	Maximum delay residual 1.4×10^{-8} , 2.0×10^{-11} , and 4.4×10^{-15} at orders 1024, 2048, 4096; redshift response exactly zero.
Static-redshift stability	Orders 512 and 1024 select the same links and agree up to the delay roundoff floor; nonlinear-lapse slope 2.0001685.
No-quotient scale separation	Exact conformal floor 1.2×10^{-18} , potential-gauge quadrature floor 2.4×10^{-10} , and smallest physical singular value 6.9×10^{-2} .

Table 5: Consolidated numerical convergence and scale-separation evidence. All values are drawn from the canonical machine-readable reports at commit a384312.

D Registered campaign records

The table below lists the per-cell medians of the registered order-1024 campaign at $K/d = 1.5$, summarized in Section 9.4 and plotted in Figure 4.

Table 6: Registered deterministic full-metric campaign at $K/d = 1.5$. Entries are medians over the five registered packet-width seeds. All geometric Jacobians were assembled at Gauss–Legendre order 1024. Spectral quantities use the full-candidate-bank parameter metric; RMSE is shown in R -normalized and raw mode coordinates. Retention is the blocked-to-unblocked $\lambda_{\min}$ ratio after separate greedy redesign under a pinned 20% angular-sector exclusion.

M	d	K	$\lambda_{\min}$	log det	κ	RMSE _w	RMSE _{raw}	worst σ_w	retention	select ms	build s
256	6	9	2.931	17.1	23.2	0.283	0.922	0.504	0.98	0.71	1.4
256	8	12	1.683	23.7	50.2	0.290	0.930	0.611	0.68	0.91	1.4
256	12	18	1.449	38.6	69.4	0.263	0.942	0.639	0.80	1.51	1.4
256	16	24	0.821	49.6	168.4	0.299	0.944	0.741	0.73	1.84	1.4
512	6	9	1.352	13.7	35.7	0.365	0.903	0.652	1.00	0.71	2.4
512	8	12	0.827	19.1	79.3	0.368	0.919	0.740	0.97	0.89	2.4
512	12	18	0.760	32.3	87.4	0.332	0.933	0.754	0.91	1.61	2.4
512	16	24	0.674	41.6	115.8	0.349	0.936	0.773	0.63	2.11	2.4
1024	6	9	0.945	9.9	30.8	0.459	0.898	0.717	1.00	0.95	5.0
1024	8	12	0.474	14.5	61.3	0.444	0.915	0.824	1.00	1.08	5.0
1024	12	18	0.477	25.8	99.5	0.395	0.928	0.823	0.99	2.07	5.0
1024	16	24	0.441	33.0	89.9	0.420	0.931	0.833	0.89	3.05	5.0
4096	6	9	0.375	3.1	18.5	0.627	0.895	0.853	1.00	1.21	20.6
4096	8	12	0.208	6.3	48.0	0.589	0.898	0.910	1.00	1.76	20.6
4096	12	18	0.206	13.3	64.1	0.529	0.919	0.910	1.01	4.83	20.6
4096	16	24	0.207	17.7	68.8	0.538	0.921	0.910	1.02	4.32	20.6

E One-shot amortized selection: full negative result

The policy repairs the soft/hard mismatch, availability masking, scale-dependent E-optimal surrogate, and numerical precision identified in the first registered attempt, and was evaluated on 100 held-out tasks.

Table 7: Held-out one-shot selector evaluation on 100 tasks. The learned DeepSets policy improves over the two originally registered weak baselines but is strictly dominated by Fisher-aware deterministic methods. Times are end-to-end per task on the same CPU target.

Method	median $\lambda_{\min}$	latency (ms)
Uniform random	0.0185	0.090
Angular spread	0.0509	0.832
DeepSets top-K	0.1202	16.56
Leverage score	0.5286	0.312
Greedy D-optimal	2.1746	0.369
Relaxed-E plus swaps	3.7157	891.69

The policy passes the original weak-baseline gate: its median paired per-task $\lambda_{\min}$ ratio is $3.57\times$ random (95% interval [2.28, 5.85]) and $1.92\times$ angular spread ([1.09, 3.26]), with 100% held-out full-rank success. It fails the registered near-oracle gate, achieving only $0.0261\times$ the relaxed-E reference ([0.0153, 0.0405]). Although it is $53.8\times$ faster than relaxed-E, a post-hoc paired analysis shows that greedy D-optimal selection is $23.3\times$ better in objective value and about $45\times$ faster. These are medians of paired taskwise ratios, not ratios of marginal medians in the table; the latter are $0.1202/0.0185 = 6.50$ for DeepSets versus random and $2.1746/0.1202 = 18.1$ for greedy versus DeepSets. The policy is therefore strictly Pareto dominated.

The leverage-score baseline reaches median $\lambda_{\min} = 0.5286$, while sequential greedy-D reaches 2.1746, a $4.1\times$ gap on the same held-out tasks. This is the cleanest non-neural evidence for the structural diagnosis: scoring candidates against a fixed full-bank matrix does not capture the conditional complementarity exposed by updating the Fisher matrix after each selection.

The original v0.2 registered outcomes are left unchanged. The large 20,000/2,000/5,000-task campaign was not run for this architecture after a documented post-hoc practical-futility decision. The result identifies a structural mismatch: the value of a candidate ray depends on the Fisher matrix of previously selected rays, while one-shot candidate scoring only weakly represents sequential complementarity. Sequential greedy selection exploits exactly that structure, in line

with classical submodular sensor-placement theory [25, 26]. Future learned designs are justified only when deterministic selection ceases to be cheap or when the design space includes correlated probes, clocks, polarization, or network topology.

F Reproducibility and data availability

The accompanying bundle contains the LaTeX manuscript, bibliography, generated figures and tables, a registration chronology, and a manifest of the committed experiment artifacts. The numerical controls incorporated here correspond to source snapshot `fdd6cfa`; full hashes and artifact checksums are listed in `REGISTRATION.md`. The density/budget table is a post-registration deterministic reduction of the already committed 320 campaign records and introduces no new forward numerics. No archival DOI is claimed in this working-paper version. A DOI-minted repository deposit should replace the development-repository reference before journal submission; until then, the versioned submission bundle is the canonical research object. Principal entry points are:

- `run_task1.sh` – seven numerical checks plus the stop-on-failure control;
- `run_static_redshift.sh` – standalone endpoint conformal-rank verification;
- `run_joint_ablation.sh` – D+R-1 and its nine numerical checks plus process control;
- `run_scaling_study.sh` – deterministic scaling and order-stability study;
- `run_full_metric_campaign.sh` – the 320-record order-1024 campaign;
- `run_weight_ablation.sh` – QFI-versus-uniform and packet-homogeneity control;
- `run_no_quotient_control.sh` – unquotiented gauge-kernel negative control;
- `analyze_density_budget.py` – deterministic reduction of candidate-density and ray-budget effects;
- `run_selector_sanity.sh`, `run_selector_eval.sh` – the frozen DeepSets negative baseline.

Machine-readable reports include:

- `task1_verification.json`, `static_redshift_experiment.json`;
- `joint_delay_redshift.json`, `scaling_study.json`;
- `full_metric_campaign.json`, `selector_evaluation.json`;
- `weight_packet_ablation.json`, `no_quotient_control.json`;

- `density_budget_analysis.json` – a post-registration reduction of the full campaign report.

The standard CI matrix runs the compact checks on macOS and Linux under Python 3.11 and 3.12; the full campaign is a separate registered script because it is longer. Numerical outputs are evidence for implementation and finite synthetic families, not substitutes for the analytic theorems.

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